

DIY Optical ToF Distance Measurement

OST – Ostschweizer Fachhochschule

CAS Sensors and Sensor Signal Conditioning

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CAS Sensors and Sensor Signal Conditioning

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Objective

Setting

OST – Ostschweizer Fachhochschule
CAS Sensors and Sensor Signal Conditioning

- Part of a CAS – Certificate Advanced Studies
 - CAS are post graduate studies (after bachelor/master)
 - CAS typically take 15 ECTS / 1 semester
 - Part time, e.g., Fri + Sat

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- CAS Sensors and Sensor Signal Conditioning^[1]
- As part of it: 5 week project work

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- Get familiar with TDCs (Time to Digital Converters)
 - Find out how much effort it takes to get TDC application running

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- First step: Electrical measurements (cable lengths)
- Second step: Optical measurements
 - Distance: 10 m
 - Resolution: 30 cm

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⇒ Focused on getting practical experience; not very scientific

GitHub

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- I'm trying not to get too technical
- Full project (HW + FW) is available on GitHub:
https://github.com/MatiMcFly/DIY_ToF

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 - Automatically translated with DeepL
- Original German version: https://github.com/MatiMcFly/DIY_ToF/blob/main/docs/documentation/documentation.pdf

Introduction

- ToF = Time of Flight
- It is the time it takes for a signal to travel from the sender to the receiver

ToF: Distance measurement

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- Role models: bat, dolphin



[2]

ToF: Distance measurement

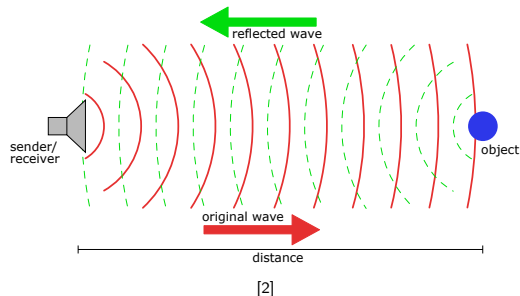
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[2]

- Technical: Ultra sonic, radar, laser



ToF: Distance measurement

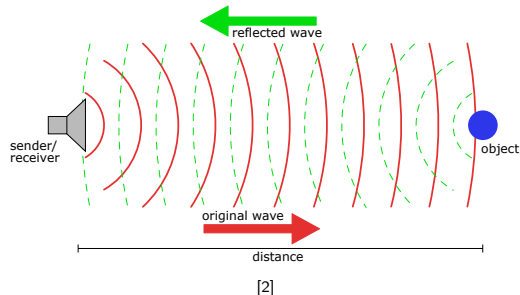
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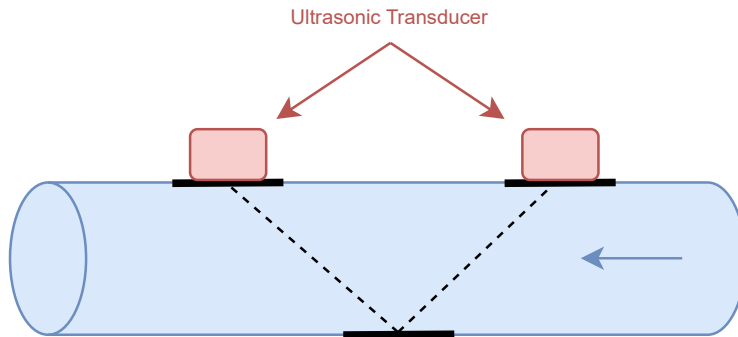


$$d = \frac{c \cdot ToF}{2}$$

ToF: Flow measurement

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- For fluids ToF can also be used to measure the flow rate (flow speed)
 - E.g., sonico NANO, sonico EDGE



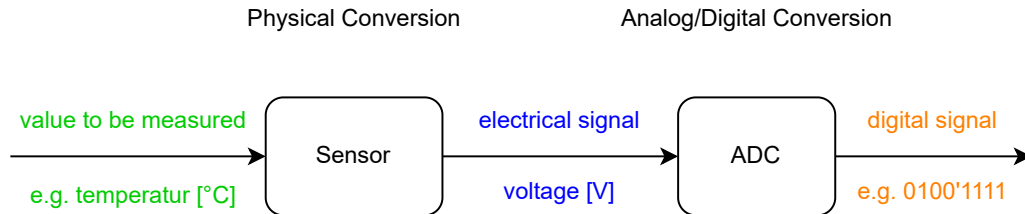
ADC

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- ADC = Analog to Digital Converter

ADC

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TDC

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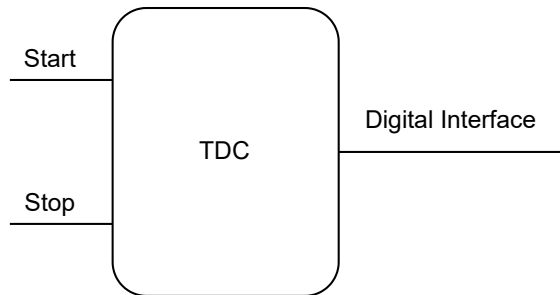
- TDC = Time to Digital Converter
- Similar to ADC, but to directly measure the time (like a stopwatch)



TDC

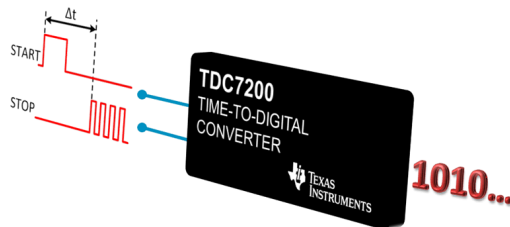
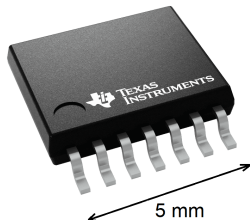
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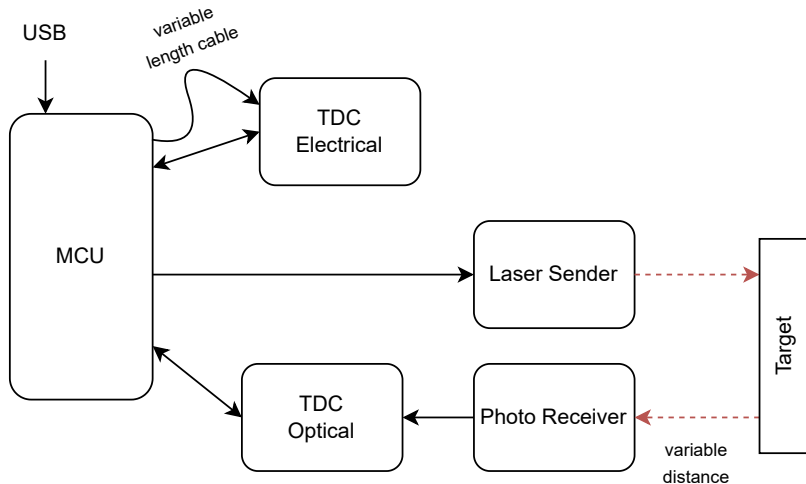
Realization

- TDC7200 from TI^[3]
 - Resolution: 55 ps
 - Range 1: 12 ns ... 500 ns
 - Range 2: 250 ns ... 8 ms
 - Digital interface: SPI



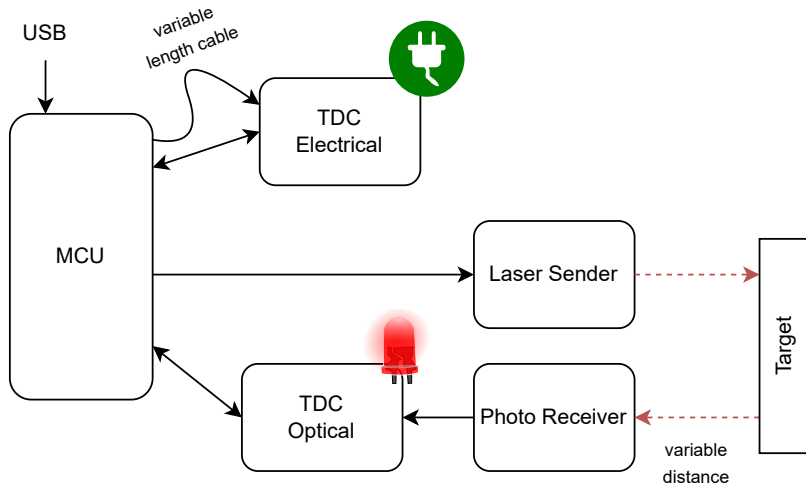
Block diagram

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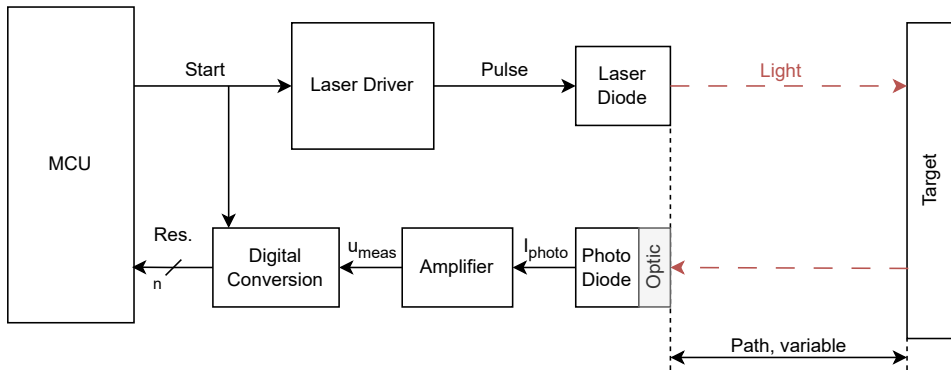
Block diagram

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Concept: Optical Part

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Realization

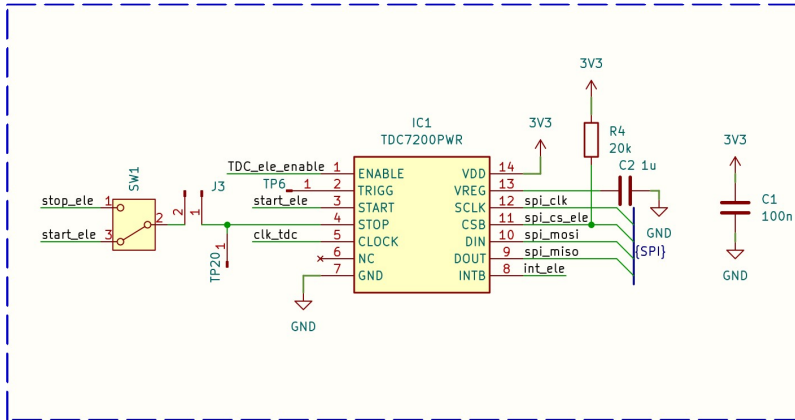
Circuits

TDC Electrical

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TDC electrical signal

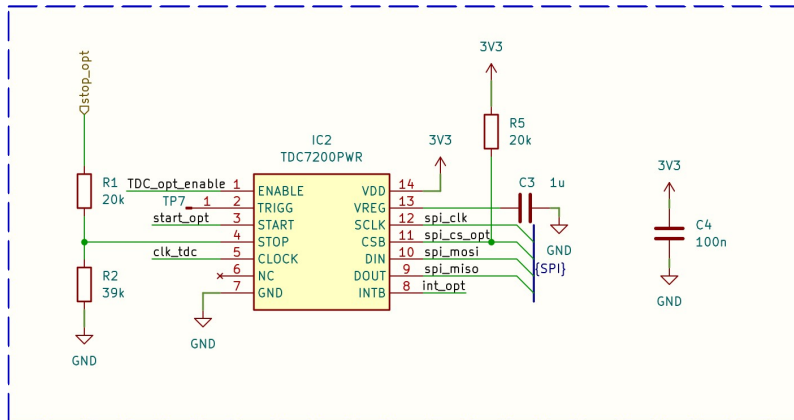


TDC Optical

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TDC optical signal



Laser Driver

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Laser driver

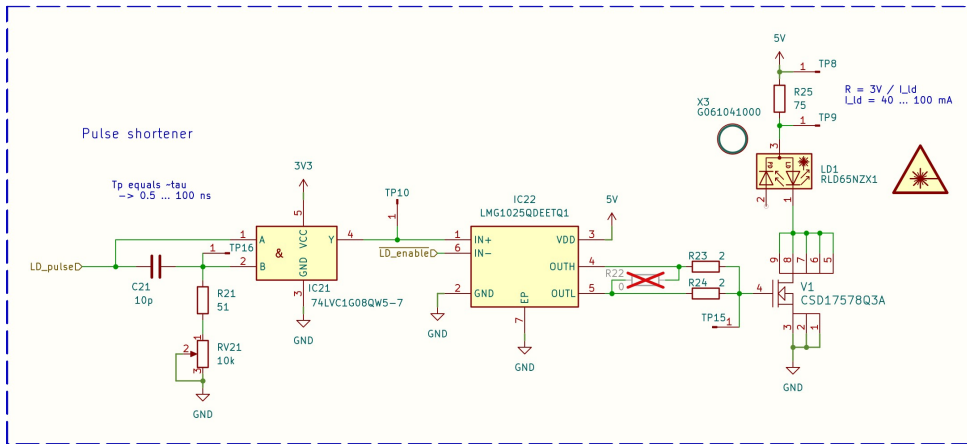
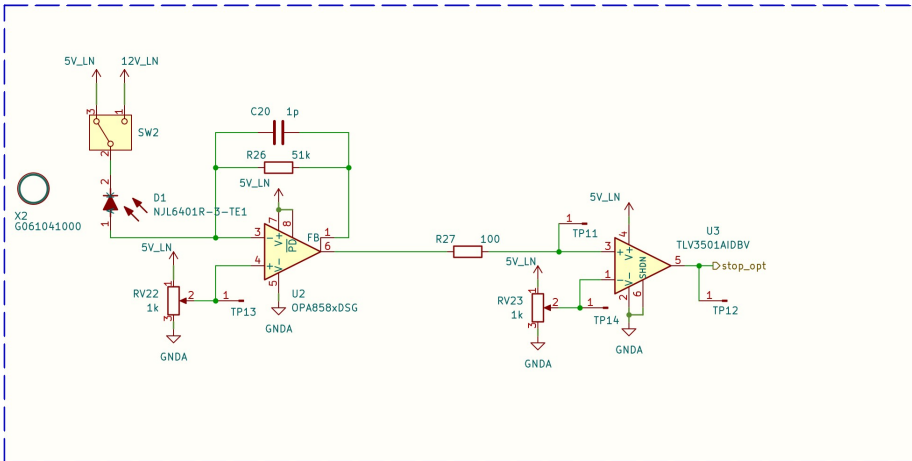


Photo Receiver

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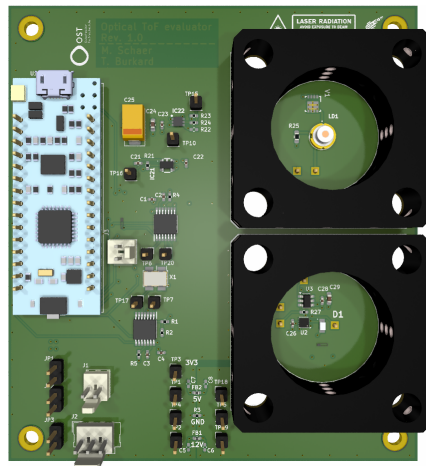


Receiver



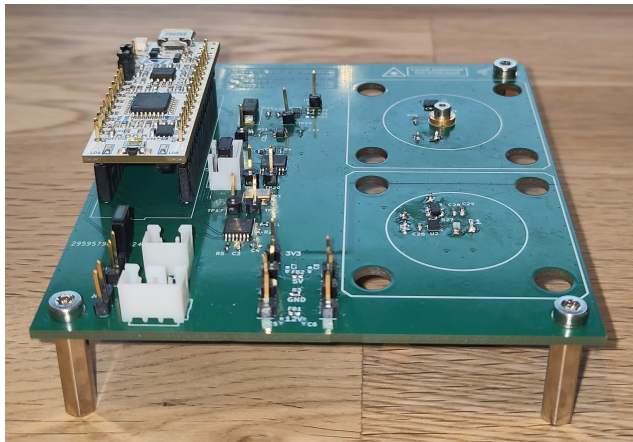
3D View

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Demonstrator

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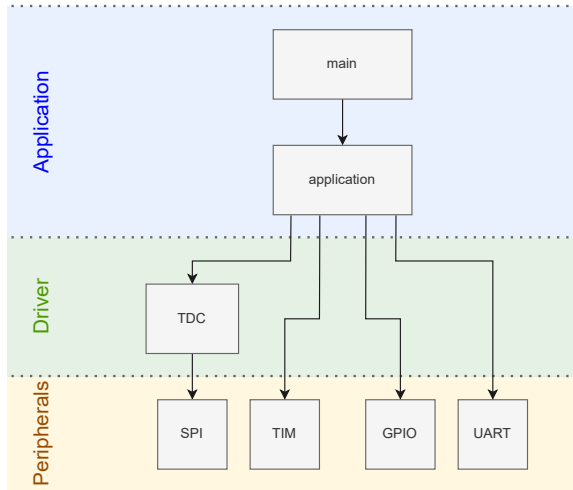


Realization

Firmware

Firmware block diagram

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Firmware example: configuration

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```
1 // Configuration
2
3 TDC_init(&tdc_ele, &hspi1, SPI_CS_ELE_GPIO, TDC_ELE_ENABLE_GPIO);
4
5 TDC_enable(&tdc_ele);
6
7 // SPI Communication
8
9 TDC_read(&tdc_ele, TDC_ADR_CONFIG1, data);
10
11 data[0] |= TDC_CONFIG1_MEAS_MODE_1;
12
13 TDC_write(&tdc_ele, TDC_ADR_CONFIG1, data);
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Firmware example: Measurement

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```
1 uint64_t tof_fs = 0;
2
3 while (1) {
4     TDC_start(&tdc_ele);
5
6     // Set Pins to High with direct register access
7     *((volatile uint32_t*)(GPIOA_BASE + 0x14)) |= (1 << 8); // START_ELE
8     *((volatile uint32_t*)(GPIOA_BASE + 0x14)) |= (1 << 11); // STOP_ELE
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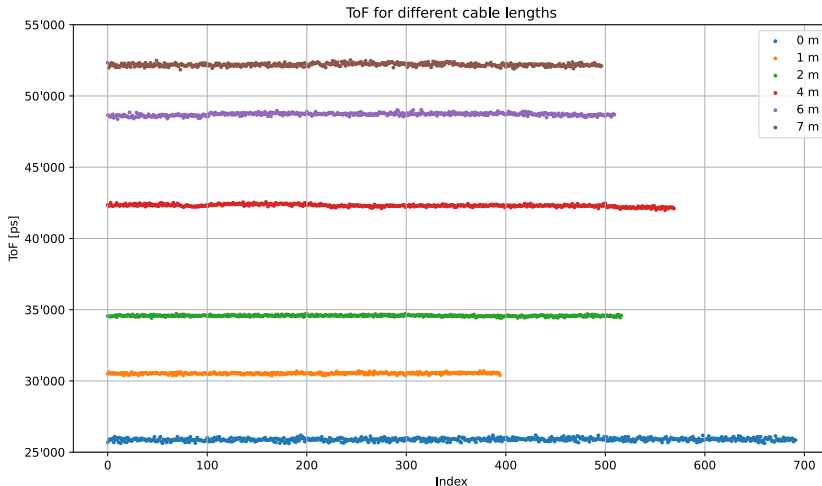
Results

Results

Electrical Results

Different cable lengths

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Different cable lengths: Results

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Length	Mean	Standard deviation	Δ at 0 m
0 m	25'890 ps	104 ps	0 ns
1 m	30'531 ps	53 ps	4.6 ns
2 m	34'575 ps	50 ps	8.7 ns
4 m	42'308 ps	88 ps	16.4 ns
6 m	48'710 ps	98 ps	22.8 ns
7 m	52'194 ps	104 ps	26.3 ns

Different cable lengths: Calculated backwards

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Length	TDC recalculated with 0.9 c
0 m	0.0 m
1 m	1.2 m
2 m	2.3 m
4 m	4.4 m
6 m	6.2 m
7 m	7.1 m

Results

Optical Results

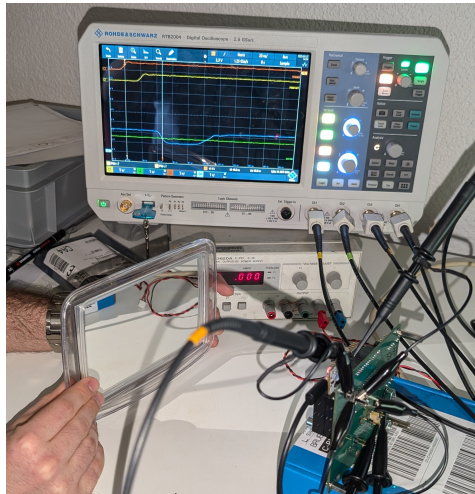
Signal Path Transmitter

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ToF measurements via mirror

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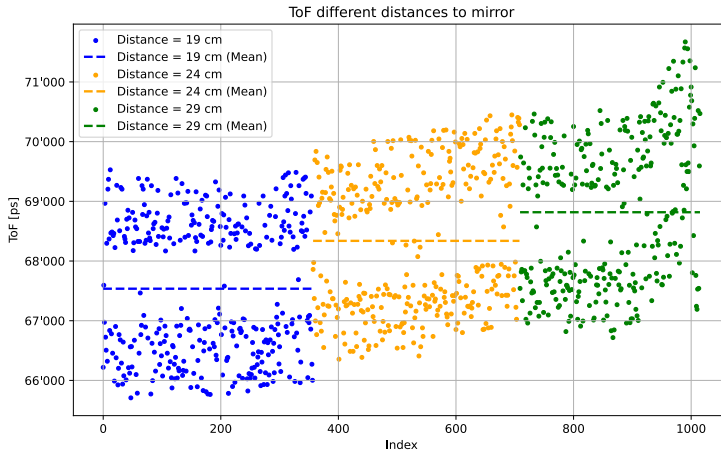
ToF measurements via mirror: DSO

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ToF measurements via mirror: Results

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ToF measurements via mirror: analysis

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Distance to mirror	Mean value	Standard deviation
19 cm	67'536 ps	1'172 ps
24 cm	68'338 ps	1'191 ps
29 cm	68'817 ps	1'273 ps

ToF measurements via mirror: analysis

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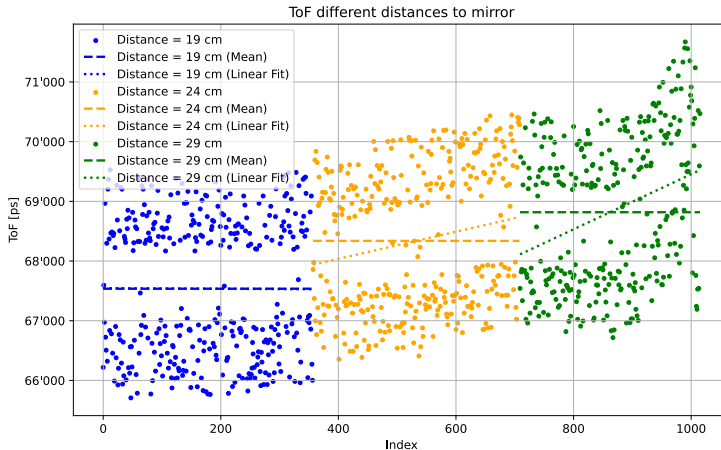
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$$\Delta ToF_{meas} \approx 0.6 \text{ ns}$$

$$\Delta ToF_{calc} = \frac{\Delta L}{c} = \frac{10 \text{ cm}}{3 \cdot 10^8 \text{ m/s}} = 0.33 \text{ ns}$$

ToF measurements via mirror: drift

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ToF measurements via mirror: >30 cm

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ToF measurements via wall

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- Unfortunately, optics not yet designed for this

Conclusion

Outlook

- Optimize the optics, try other focal lengths
- Minimize jitter (higher clock or external clock)
- Switch GPIOs faster
- Optical receive path
 - Different amplifications TIA
 - Different switching thresholds for comparator
- Optical transmission path: try higher transmission power
- Analysis of optical results: why two clusters?
- Mechanical test setup for mirror

Lessons Learned

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- Overall very satisfied
 - Learned a lot of new things
 - Unfortunately, a little short of time
 - Time resolution less of a problem than the amount of light received
- ⇒ Invest more time in optics and optoelectronics
- ⇒ Easy to get an application running with a TDC7200

Demonstration

Questions?

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References

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- [1] OST. CAS Sensors and Sensor Signal Conditioning, 2025. URL <https://www.ost.ch/de/weiterbildung/weiterbildungsangebot/technik/mikroelektronik-und-sensorik/cas-sensorik-und-sensor-signal-conditioning>. visited 2025-03-25.
- [2] Wikipedia. Laufzeitmessung, 2023. URL <https://de.wikipedia.org/wiki/Laufzeitmessung>. visited 2023-02-10.
- [3] TI. TDC7200, 2025. URL <https://www.ti.com/product/TDC7200>. visited 2025-03-25.
- [4] TI. TDC7200 Datasheet, 2016. URL <https://www.ti.com/lit/ds/symlink/tdc7200.pdf?ts=1742823831671>. visited 2025-03-25.